

A Literal Fixed-Atom Candidate-Mechanism Gate: Formula Completeness and Scoped Theorem Exhaustion

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology

July 2026

Abstract

We audit a declared seven-source theorem corpus against the literal fixed-atom determinant-two two-Möbius target selected by MVP9. Two theorems act directly on the arithmetic coefficient: the Teräväinen–Walker logarithmically averaged affine twist at every fixed atom, and the Tao–Teräväinen almost-scale correlation theorem after periodic reassembly at a rational atom. Neither preserves the full conjunction of natural normalization, deterministic all-prefix/all-scale quantifiers, fixed- X power, and the actual-support axis. Thus the declared corpus has two direct candidates and zero eligible mechanisms.

The audit also finds a prior formula-level gap. The written direct transform uses the terminal block $N < t(z) \leq 2N$, while the bad-endpoint contract uses the cumulative prefix $0 < t(z) \leq T$. The source-locked direct contract does not specify a cumulative replacement, so the claimed specialization $N = T$ is not formula-certified. The named atom, packet schedule, uniform constant, exponent, ranges and physical loss ledger are also absent. The result is a fail-closed L1 scoped exhaustion, not an L2 estimate or a global nonexistence theorem. The batch therefore stops at TPC-193 for user confirmation.

1 Gate scope and literal coefficient

Put $e(x) = \exp(2\pi ix)$. TPC-167 writes the actual arithmetic core as [13]

$$a, s \geq 1, \quad d, u \in \mathbb{Z}, \quad (a, s) = 1, \quad as \text{ odd}, \quad su - ad = 2, \quad (1)$$

$$q = as, \quad t(z) = ad + qz, \quad c_z = \mu(d + sz)\mu(u + az). \quad (2)$$

The determinant identity also gives

$$t(z) = a(d + sz), \quad t(z) + 2 = s(u + az),$$

so (2) is the literal fixed- $h_0 = 2$ two-Möbius coefficient, not a majorant or synthetic surrogate.

The MVP9 eligibility signature is [22]

$$\begin{aligned} &(\text{ACTUAL_FIXED_H0_PACKET}, \text{NAMED_FIXED_ATOM}, \\ &\text{DETERMINISTIC_ALL_PREFIX}, \text{DETERMINISTIC_ALL_SCALE}, \\ &\text{FIXED_X_POWER_FIXED_ATOM}, \text{ACTUAL_ACTIVE_SUPPORT}). \end{aligned} \quad (3)$$

An eligible mechanism must be theorem-backed, act directly on (2), preserve all six coordinates, and state its constants, ranges, normalization and losses. Phase L^2 , almost-every phase, size-only control and renamed forms of the three stopped method cells are excluded before promotion.

2 The formula-completeness gate

2.1 Two source-locked summation domains

The only explicit direct transform in the locked chain is

$$I_N^{\text{blk}} = \{z : N < t(z) \leq 2N\}, \quad F_N^{\text{blk}}(\alpha) = \frac{q}{N} \sum_{z \in I_N^{\text{blk}}} c_z e(-\alpha z). \quad (4)$$

The TPC-149 arithmetic theorem uses the same terminal block [10]. In contrast, TPC-159 writes the exact cumulative domain and q/T normalization for $c_z \rho(z)$, while TPC-184 freezes the cumulative actual-core object as the prescribed-atom bad-endpoint contract. Combining that domain with the TPC-167 additive-character convention, the audited comparison target is

$$B_T(\alpha) = \frac{q}{T} \sum_{\substack{z \in \mathbb{Z} \\ 0 < t(z) \leq T}} c_z e(-\alpha z), \quad (5)$$

uniformly over prescribed prefix endpoints [12, 13, 18].

TPC-183 states that a direct estimate specializes to the bad-endpoint estimate by setting $N = T$, while TPC-189 describes the direct target in words as “all prefix” [17, 20]. Neither paper writes a direct summation domain or a prefix index set. If “direct” retains (4), setting $N = T$ leaves $T < t(z) \leq 2T$ and does not produce (5). If it instead means a new cumulative direct prefix, that new formula and its admissible range are absent.

Proposition 2.1 (Formula-level fail-closed boundary). *The source-locked record does not uniquely instantiate the direct target’s summation domain. Consequently the TPC-183 implication*

$$\text{direct fixed atom} \implies \text{bad-endpoint fixed atom}$$

is not formula-certified by the locked formulas. This audit records the implication as unresolved, not false.

Proof. Equations (4) and (5) have different domains. The substitution $N = T$ does not identify them. No locked direct-target formula supplies a third definition that resolves the difference. $\square$

2.2 Missing physical fields

TPC-180 finds zero value-bearing named-phase records and zero production packet-coordinate rows: the atom identifier, phase value modulo one, source locator and packet-schedule locator are null [15]. TPC-189 also does not freeze the admissible X, N, q ranges, a positive pointwise exponent σ , the uniform constant C , or a complete physical loss ledger. Thus the desired skeleton

$$|F_N(\alpha_\star)| \leq C X^{-\sigma}$$

cannot yet be parsed into a unique literal theorem call. Fixed $h_0 = 2$ remains a data fact only.

3 Declared primary-source corpus

The corpus is finite and explicit. It contains the five primary sources already cited by the direct arithmetic lineage and two additional sources found by an exact twisted-correlation/Fourier-uniformity scan:

Source	Native theorem cell	Gate disposition
Matomäki–Radziwiłł–Tao (2015)	shift-averaged Chowla; single-factor short Fourier sums	averages the shift or interval origin; no fixed determinant-two two-factor theorem [2].
Tao (2016)	logarithmic two-point Elliott/Chowla	fixed affine and reciprocal weight; subsumed for the twisted gate by Teräväinen–Walker [5].
Helfgott–Radziwiłł (2021)	quantitative shift-one parity result	TPC-138 proves the even/determinant-one transfer misses the active odd determinant-two carrier [1, 9].
Pilatte (2023)	improved two-point logarithmic Chowla bounds	almost-scale shift-one input; the same carrier firewall applies [3, 9].
Tao–Teräväinen (2026)	quantitative natural correlations outside exceptional scales	yields the rational-periodic direct candidate below [6].
Teräväinen–Walker (2025)	logarithmic affine correlations twisted by every additive character	yields the strongest all-fixed-atom direct candidate below [7].
Pilatte (2026)	single-factor Fourier uniformity averaged over interval origins	multiplication by a second affine Möbius factor is not a theorem-backed operation [4].

All seven records are reviewed and mapped. Exhaustion below is therefore closed only inside this declared corpus; it says nothing about all possible future literature or mechanisms.

4 The two direct theorem candidates

4.1 Every fixed atom under logarithmic averaging

Teräväinen–Walker Lemma 4.2(1) states that, for fixed nonparallel positive affine forms, real-valued bounded multiplicative functions with the first nonpretentious, and every $\gamma \in \mathbb{R}$,

$$\lim_{X \rightarrow \infty} \frac{1}{\log X} \sum_{n \leq X} \frac{f_1(a_1 n + h_1) f_2(a_2 n + h_2) e(\gamma n)}{n} = 0. \quad (6)$$

Take $f_1 = f_2 = \mu$. Choose a fixed z_0 for which both translated intercepts are positive, set $z = n + z_0$, and use

$$(a_1, h_1) = (s, d + sz_0), \quad (a_2, h_2) = (a, u + az_0), \quad \gamma = -\alpha_\star.$$

Then the nonparallel determinant is

$$s(u + az_0) - a(d + sz_0) = su - ad = 2,$$

so (6) acts directly on (2) at every fixed prescribed atom. The translation changes the phase only by the unit scalar $e(-\alpha_\star z_0)$.

This is the strongest direct candidate in the corpus. Its native normalization is nevertheless reciprocal/logarithmic, its data are fixed, and its conclusion is qualitative $o(1)$. TPC-129 proves that terminal reciprocal cancellation has no formal implication to an ordinary all-prefix sum without extra path information [8]. Hence (6) supplies neither natural q/N control nor a uniform fixed- X power.

4.2 A rational atom outside exceptional scales

Tao–Teräväinen Theorem 3.1, after the quotient lift and periodic reassembly of TPC-149, gives absolute $\eta_0, \kappa_0 > 0$ and $\mathcal{E}_X^\star \subset [\sqrt{X}, X]$ with

$$\frac{1}{\log X} \int_{\mathcal{E}_X^\star} \frac{dt}{t} \ll (\log X)^{-\kappa_0}$$

such that

$$\frac{q}{N} \left| \sum_{N < t(z) \leq 2N} c_z \rho(z) \right| \ll \|\rho\|_\infty (\log X)^{-\kappa_0} \quad (7)$$

whenever $qR \leq (\log X)^{\eta_0}$, $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X^*$, and ρ has period R . For a rational atom $\alpha_* = r/R$, put $\rho(z) = e(-rz/R)$. This preserves the literal coefficient and the natural q/N normalization [10, 11].

The production record, however, contains no source-backed rational atom or denominator. Even if it did, (7) controls one dyadic block only, deletes exceptional scales, and saves a power of $\log X$, whose fixed- X exponent is zero. TPC-159 and TPC-170 leave respectively a dyadic bad shadow or an almost-every-phase, prescribed-schedule quantifier; neither repairs all required axes [12, 14].

5 Scoped exhaustion theorem

Theorem 5.1 (TPC-193 candidate-mechanism gate). *In the declared seven-source corpus:*

$$\#\{\text{primary records}\} = 7, \quad \#\{\text{direct candidates}\} = 2, \quad \#\{\text{eligible candidates}\} = 0.$$

The target contract is NOT-TESTABLE at the formula level, with first missing field

DIRECT_TARGET_SUMMATION_DOMAIN_AND_PREFIX_INDEX.

Independently, each direct candidate fails required analytic axes even under either possible domain interpretation. No theorem-backed genuine sublemma is available to determine TPC-194.

Proof. Proposition 2.1 proves target-formula incompleteness. For (6), the first native mismatch is reciprocal/logarithmic rather than natural endpoint normalization; fixed data, all-prefix uniformity, all-scale uniformity, a fixed- X rate and actual support also fail. For (7), the atom has no rational source record and the theorem has an exceptional-scale set, a terminal-block domain and only logarithmic saving. Thus neither candidate satisfies the eligibility conjunction. The other five primary sources fail the direct two-factor prefilter. The machine registry reviews all seven records and has no unmapped or unreviewed cell. $\square$

6 Scoped stop, endpoint ledger and claim firewall

The exact stopped cell is

TPC193_DECLARED_CANDIDATE_MECHANISM_CORPUS_V1.

It is a workflow stop for this batch and a corpus-scoped exhaustion. It does not stop either O161 pointwise theorem, the global architecture, or mathematical search outside the declared corpus. The three older method cells remain separately scoped [16, 19, 21]:

phase_metric_uncontrolled_atomic,
SIZE_ONLY_LOCAL_OSCILLATION_METHOD,
PARSEVAL_CHEBYSHEV_TO_PRESCRIBED_ATOM.

The endpoint ledger remains

$$\sigma_{\text{named atom}} = 0, \quad \text{strict } 1/400 = \text{UNPAID}.$$

There is no program-positive L2, fixed-atom decay, prime-pair lower bound, or twin-prime theorem. Hashes certify integrity only, not theorem truth. Following the dynamic route rule, the next action is USER_CONFIRMATION_REQUIRED; TPC-194 is not created.

7 Reproducible audit

The adjacent generator reconstructs the payload from source-locked repository artifacts, validates all source hashes, recomputes candidate eligibility, executes deep-copy mutation attacks against semantic and schema validators, and emits exact Draft 2020-12 schemas. Every object is closed, every leaf is constant-locked, arrays have exact prefix schemas, duplicate JSON keys and nonfinite numbers are rejected, and `-check` performs canonical byte comparison of the payload, audit and both schemas.

References

- [1] Harald Andrés Helfgott and Maksym Radziwiłł. Expansion, Divisibility and Parity, 2021.
- [2] Kaisa Matomäki, Maksym Radziwiłł, and Terence Tao. An Averaged Form of Chowla’s Conjecture. *Algebra & Number Theory*, 9(9):2167–2196, 2015. doi: 10.2140/ant.2015.9.2167.
- [3] Cédric Pilatte. Improved Bounds for the Two-Point Logarithmic Chowla Conjecture, 2023.
- [4] Cédric Pilatte. Improved Bounds for the Fourier Uniformity Conjecture, 2026.
- [5] Terence Tao. The Logarithmically Averaged Chowla and Elliott Conjectures for Two-Point Correlations. *Forum of Mathematics, Pi*, 4:e8, 2016. doi: 10.1017/fmp.2016.6.
- [6] Terence Tao and Joni Teräväinen. Quantitative Correlations and Some Problems on Prime Factors of Consecutive Integers, 2026.
- [7] Joni Teräväinen and Aled Walker. On a Bohr Set Analogue of Chowla’s Conjecture. *Mathematische Zeitschrift*, 310(4):75, 2025. doi: 10.1007/s00209-025-03770-2.
- [8] Liang Wang. A Log-Chowla Quantifier Firewall. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-129-log-chowla-quantifier-firewall, 2026.
- [9] Liang Wang. Quantitative Shift-One Shadows of the Determinant-Two Frame. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-138-even-carrier-shift-one-firewall, 2026.
- [10] Liang Wang. A Small-Polylogarithmic Determinant-Two Möbius Corridor. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-149-small-polylog-determinant-two-mobius-corridor, 2026.
- [11] Liang Wang. An Exact Major-Arc Gate for Additive Phases. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-158-additive-phase-major-minor-gate, 2026.
- [12] Liang Wang. Dyadic-Shadow Lifting from Good-Scale Möbius Correlations to Almost-Endpoint Prefixes. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-159-dyadic-shadow-prefix-lifting, 2026.
- [13] Liang Wang. A Parseval Corridor for Direct Additive Twists. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-167-direct-additive-twist-parseval, 2026.
- [14] Liang Wang. Metric All-Prefix Return on Prescribed Determinant-Two Packet Corridors. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-170-metric-packet-corridor-return, 2026.
- [15] Liang Wang. A Source-Locked Census for the Named Physical Phase. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-180-production-phase-registry-census, 2026.
- [16] Liang Wang. The Metric-to-Fixed-Atom Selector Gate. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-181-metric-fixed-atom-selector-gate, 2026.
- [17] Liang Wang. The Two Pointwise Fixed-Atom Parents. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-183-pointwise-parent-interface-comparison, 2026.
- [18] Liang Wang. A Literal Contract for the Bad-Endpoint Pointwise Target. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-184-bad-endpoint-literal-target-contract, 2026.

- [19] Liang Wang. A Sharp Barrier for Size-Only Control of Local Oscillation. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-187-size-only-local-oscillation-barrier, 2026.
- [20] Liang Wang. A Literal Contract for the Direct Additive-Twist Target. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-189-direct-twist-literal-target-contract, 2026.
- [21] Liang Wang. Why Parseval Does Not Evaluate a Prescribed Phase Atom. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-190-parseval-to-atom-method-barrier, 2026.
- [22] Liang Wang. MVP9 after the Pointwise Fixed-Atom Frontier Audit. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-192-mvp9-pointwise-frontier-route-decision, 2026.